

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Comparative Analysis

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Comparative Analysis

1. Apply **SSL/TLS and IPsec protocols** to a secure communication scenario and determine which provides better security for web-based and network-layer protection.
2. Design a secure email system by comparing **end-to-end encryption and gateway-based encryption**, and justify the most suitable approach for enterprise use.
3. Apply **HMAC and Digital Signature techniques** to a data transmission scenario and analyze which method provides better authentication and integrity.
4. Create a **spam detection system** by comparing supervised learning and unsupervised learning approaches, and evaluate their effectiveness in real-world email filtering.
5. Design a secure communication framework by comparing **AES and RSA encryption techniques**, and determine their roles in achieving confidentiality and efficiency.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand